Lab Procedure
Light Resistors

Introduction

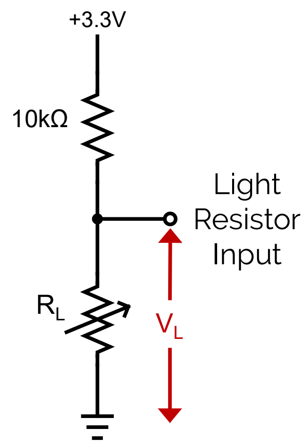
Ensure the following:

1. You have reviewed the Application Guide – Light Resistors
2. Make sure the **Mechatronic Sensors Trainer** is out of its case; it is powered using the provided USB cable connected to the PC.
   a. The LED next to the USB C port will stay white after the touch screen starts loading.
   b. The load screen with the Quanser Logo should disappear after a few seconds and you should see some evenly spaced crosses on a black background on the touch screen.
3. You have a sheet of white paper.

Exploring Light Resistors

1. Launch Visual Studio Code or your preferred Python IDE. If using Visual Studio Code, click on **File** > **Open Folder**…, and browse to the directory containing the python files and lab procedure for the Light Resistors lab (…/sensors_trainer/1_fundamentals /light_resistor/hardware/python). Open this directory. Open **light_resistor.py**.
2. Run the script by clicking the play button ▷ at the top right corner of the code editor. This will open a **Light Resistor** scope displaying the voltage reading from the light resistor, and the stored minimum and maximum readings from the sensor.
3. Place a dark colored object 5cm in front of the light resistor on the device. Using the terminal output, what is the value read by the light resistor?
4. Move the object closer to or farther away from the sensor. How does the signal change as you move the object? Using the values printed to the terminal, what are the highest and lowest signal values that are read by the sensor?
5. Stop the script by clicking back on the terminal in Visual Studio Code, then pressing **Ctrl + C**, and then clicking enter when prompted in the terminal.
6. Repeat steps 2-5 with a sheet of white paper. How does the signal change as you move the sheet of paper closer or farther away?

7. Consider the diagram below, which shows the internal circuit in the Sensors Trainer for the light resistor input. The circuit is a voltage divider made up of an $10k\Omega$ resistor connected to a 3.3 V voltage source and the light resistor, which is connected to ground. The light resistor input measures the voltage V_L, across the light resistor. Using this diagram, write an equation for the voltage V_L across the light resistor as a function of its resistance R_L. Use this equation to explain how the voltage across the light resistor changes with the amount of light detected. Does the resistance increase or decrease when more light is detected?



8. Open the datasheet for the light resistor, located in the directory .../ sensors_trainer/1_fundamentals/light_resistor. Referring to the datasheet, verify if the resistance increases or decreases when more light is detected. Report the value of the dark resistance from the datasheet. This is the nominal resistance when the least amount of light is detected. Use your equation from step 7 to find the expected voltage at the dark resistance. Run your script and report the measured voltage printed to the terminal when you completely cover the square cutout of the light resistor with the dark colored object. How does this value compare to the expected voltage?

9. Repeat step 6 in different lighting environments, such as a room with brighter or dimmer lights, or outside. How does this affect the sensitivity of the light sensor reading?

10. Stop, save and close your program.

11. Power OFF the Mechatronic Sensors Trainer by disconnecting its USB C cable, disconnect any external sensors and pack them along the base and the USB cables in the provided case.